

AN INTERRUPTED TENNIS GAME

It is not uncommon for someone watching a tennis game on television to be bombarded by advertisements for funds that did (until that minute) outperform others by some percentage over some period. But, again, why would anybody advertise if he didn't happen to outperform the market? There is a high probability of the investment coming to you if its success is caused entirely by randomness. This phenomenon is what economists and insurance people call adverse selection. Judging an investment that comes to you requires more stringent standards than judging an investment you seek, owing to such selection bias. For example, by going to a cohort composed of 10,000 managers, I have 2/100 chances of finding a spurious survivor. By staying home and answering my door bell, the chance of the soliciting party being a spurious survivor is closer to 100%.

THE BIRTHDAY PARADOX

The most intuitive way to describe the data mining problem to a non-statistician is through what is called the birthday paradox, though it is not really a paradox, simply a perceptual oddity. If you meet someone randomly, there is a one in 365.25 chance of your sharing their birthday, and a considerably smaller one of having the exact birthday of the same year. So, sharing the same birthday would be a coincidental event that you would discuss at the dinner table. Now let us look at a situation where there are 23 people in a room. What is the chance of there being two people with the same birthday? About 50%. For we are not specifying which people need to share a birthday; any pair works.

IT'S A SMALL WORLD!

A similar misconception of probabilities arises from the random encounters one may have with relatives or friends in highly unexpected places. "It's a small world" is often uttered with surprise. But these are

not improbable occurrences – the world is much larger than we think. It is just that we are not truly testing for the odds of having an encounter with one specific person, in a specific location at a specific time. Rather, we are simply testing for any encounter, with any person we have ever met in the past, and in any place we will visit during the period concerned. The probability of the latter is considerably higher, perhaps several thousand times the magnitude of the former.

When the statistician looks at the data *to test a given relationship*, say to ferret out the correlation between the occurrence of a given event, like a political announcement, and stock market volatility, odds are that the results can be taken seriously. But when one throws the computer at data, looking for just about *any* relationship, it is certain that a spurious connection will emerge, such as the fate of the stock market being linked to the length of women's skirts. And just like the birthday coincidences, it will amaze people.

DATA MINING, STATISTICS, AND CHARLATANISM

What is your probability of winning the New Jersey lottery twice? One in 17 trillion. Yet it happened to Evelyn Adams, whom the reader might guess should feel particularly chosen by destiny. Using the method we developed above, Harvard's Percy Diaconis and Frederick Mosteller estimated at 30 to 1 the probability that someone, somewhere, in a totally unspecified way, gets so lucky!

Some people carry their data mining activities into theology – after all, ancient Mediterraneans used to read potent messages in the entrails of birds. An interesting extension of data mining into biblical exegesis is provided in *The Bible Code* by Michael Drosnin. Drosnin, a former journalist (seemingly innocent of any training in statistics), aided by the works of a "mathematician", helped "predict" Rabin's assassination by deciphering a bible code. He informed Rabin, who obviously did not take it too seriously. *The Bible Code* finds statistical irregularities in the Bible; these help predict some such events. Needless to say that the book sold well.